

Vipul Harsh

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Interests	Current interests: Failure diagnosis in networked systems, reliability of AI agents, network topology Broad interests: Networked and Distributed Systems, ML for Systems, Algorithms	
Experience	Postdoctoral researcher, Conviva, Foster City, California	July 2024 - Ongoing
	Visiting researcher, CMU Advisors: Vyas Sekar , Hui Zhang	
	Research intern, VMware Research, Palo Alto, California (remote) Advisors: Sujata Banerjee , Radhika N. Mysore	June 2020 - Sept. 2020
Education	Software engineering intern, Google, Sunnyvale, California Advisors: Gautam Kumar , Nandita Dukkupati	May 2018 - August 2018
	University of Illinois at Urbana-Champaign Ph.D., Computer Science Advisor: P. Brighten Godfrey	2017 - 2024
	University of Illinois at Urbana-Champaign M.S., Computer Science Advisor: Laxmikant Kale	2015 - 2017
Publications	Indian Institute of Technology, Bombay B.Tech. (Honors), Computer Science and Engineering	2011 - 2015
	MoCE: A Mixture of Context Aware Experts Framework for Troubleshooting Internet-scale Services Vipul Harsh , Sayan Sinha, Henry Milner, B. Aditya Prakash, Vyas Sekar, Hui Zhang NSDI 2026	
	Starfish: A Topology-Routing Co-Design for Small-Scale Data Centers Ancheng Zhou, Vipul Harsh , Sangeetha A. Jyothi, Maria Apostolaki, P. Brighten Godfrey NSDI 2026	
Publications	Automatically Surfacing Opportunities for Improvements In Internet-Scale Applications Vipul Harsh , Sayan Sinha, Henry Milner, Haijie Wu, B. Aditya Prakash, Vyas Sekar, Hui Zhang HotNets 2025	
	TraceWeaver: Distributed Request Tracing for Microservices Without Application Modification Sachin Ashok, Vipul Harsh , P. Brighten Godfrey, Radhika Mittal, Srinivasan Parthasarathy, Larisa Shwartz SIGCOMM 2024	
	Murphy: Performance Diagnosis of Distributed Cloud Applications Vipul Harsh , Wenxuan Zhou, Sachin Ashok, Radhika N. Mysore, P. Brighten Godfrey, Sujata Banerjee SIGCOMM 2023	
Publications	• VMware's blog post about product adoption: https://shorturl.at/efvT2	
	Flock: Accurate Datacenter Fault Localization at Scale Vipul Harsh , Tong Meng, Kapil Agrawal, P. Brighten Godfrey CoNEXT 2023	
	Optimal Round and Sample-Size Complexity for Partitioning in Parallel Sorting Wentao Yang*, Vipul Harsh* , Edgar Solomonik (*: equal contribution) SPAA 2023	
Publications	Spineless Datacenters Vipul Harsh , Sangeetha A. Jyothi, P. Brighten Godfrey HotNets 2020	

Histogram Sort with Sampling

Vipul Harsh, Laxmikant Kale, Edgar Solomonik
SPAA 2019

Manuscripts in preparation	There's Waldo: Localizing failures among symmetric components Vipul Harsh, Rahul Bothra, P. Brighten Godfrey
Patents	• On-demand Network Incident Graph Generation (US Patent App. 18/094,378) Vipul Harsh, Wenxuan Zhou, Radhika Niranjana Mysore, Philip Brighten Godfrey, Sujata Banerjee • Network Incident Root-Cause Analysis (US Patent App. 18/094,379) Vipul Harsh, Wenxuan Zhou, Radhika Niranjana Mysore, Philip Brighten Godfrey, Sujata Banerjee • Providing Explanation of Network Incident Root Causes (US Patent App. 18/094,380) Vipul Harsh, Wenxuan Zhou, Radhika Niranjana Mysore, Philip Brighten Godfrey, Sujata Banerjee
Awards	• Selected for the NSF-NetS early career workshop, January 2025 at Alexandria, VA, USA • Represented IIT Bombay at the ACM ICPC World Finals 2015. Highest ranked team from India • Ranked 49 in IIT-JEE 2011, amongst 500,000 candidates • Rank 1 in International Mathematics Olympiad, 2009 conducted by Science Olympiad Foundation
Internships	• Research Internship, UC Berkeley June 2019 - August 2019 Distributed garbage collection for actor-based systems • Research Internship, Georgia Tech May 2014 - July 2014 Large scale simulations for polydisperse hydrodynamic particle systems • Research Internship, LaBRI, France May 2013 - July 2013 Algorithms for computing coverability sets of Petri Nets
Teaching	• Teaching Assistant, Cloud Networking, UIUC (Spring 2024) • Teaching Assistant, Probability and Statistics for Computer Science, UIUC (Spring 2022) • Teaching Assistant, Discrete Mathematics, IIT Bombay • Teaching Assistant, GPU Programming and Applications Workshop (GPA), IIT Bombay
Talks	• HotNets 2025, College Park, Maryland November 2025 Automatically Surfacing Opportunities for Improvements In Internet-Scale Applications • CMU, Pittsburgh, Pennsylvania November 2025 Abstractions for high-coverage, extensible and scalable root cause analysis (RCA) Hosted by Prof. Vyas Sekar • MIT, Cambridge, Massachusetts July 2024 Failure Diagnosis in Networked systems Hosted by Prof. Christina Delimitrou • Conviva, Foster city, California March 2024 Failure Diagnosis in Networked systems Hosted by Prof. Vyas Sekar • CoNEXT 2023, Paris December 2023 Flock: Accurate Network Fault Localization at Scale • SIGCOMM 2023, New York City September 2023 Murphy: Performance Diagnosis of Distributed Cloud Applications • SPAA 2023, Orlando, Florida June 2023 Optimal Round and Sample-Size Complexity for Partitioning in Parallel Sorting • EnvoyCon at KubeCon, Detroit, Michigan October 2022 Distributed Tracing without the pain • VMware Research, Palo Alto, California August 2022 Murphy: Performance diagnosis of Distributed Cloud Applications • VMware RADIO, San Francisco, California May 2022 Murphy: Performance diagnosis of Distributed Cloud Applications

- **HotNets 2020**, Chicago, Illinois
Spineless Data Centers November 2020
- **VMware**, Palo Alto, California February 2020
Fast and accurate datacenter fault localization
- **Google**, Sunnyvale, California August 2019
Fast and accurate datacenter fault localization
- **SPAA 2019**, Phoenix, Arizona June 2019
Histogram sort with sampling
- **Charm++ Workshop**, Urbana, Illinois April 2017
Histogram sort with sampling

References
(on request)

- **Vyas Sekar**, CMU, Conviva
- **Hui Zhang**, CMU, Conviva
- **Brighten Godfrey**, UIUC
- **Sujata Banerjee**, Microsoft Research
- **Edgar Solomonik**, UIUC

Service

- Program Committee, Conext 2025